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LIQUID DISCHARGE APPARATUS AND LIQUID DISCHARGING METHOD

Abstract

A liquid discharge apparatus includes a liquid chamber including a nozzle plate having a nozzle hole, a supply unit to supply a liquid to the liquid chamber, a valve disposed in the liquid chamber, and a valve conveyor to move the valve. When the liquid is to be discharged and when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole. An amount of the liquid to be supplied to the nozzle hole when the liquid is not to be discharged is smaller than an amount of the liquid to be supplied to the nozzle hole when the liquid is to be discharged.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This patent application is based on and claims priority pursuant to 35 U.S.C. § 119 (a) to Japanese Patent Application No. 2024-036999, filed on Mar. 11, 2024, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a liquid discharge apparatus and a method of discharging liquid.

Background Art

[0003] In liquid discharge apparatuses in which valves open and close, pressurized liquid that is supplied to a liquid chamber is discharged as the nozzle holes that are formed on the liquid chamber are opened by the movement of valves, and the liquid is not discharged when the nozzle holes are closed by the movement of the valves.

[0004] In order to reduce the ejection abnormality, for example, vibration is given to the liquid levels inside the nozzle holes when liquid is not to be discharged. By so doing, the thickened liquid is stirred in the liquid chamber.

SUMMARY

[0005] The present disclosure described herein provides a liquid discharge apparatus and a method of discharging a liquid. The liquid discharge apparatus includes a liquid chamber including a nozzle plate having a nozzle hole, a supply unit to supply a liquid to the liquid chamber, a valve disposed in the liquid chamber, and a valve conveyor to move the valve. When the liquid is to be discharged and when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole. An amount of the liquid to be supplied to the nozzle hole when the liquid is not to be discharged is smaller than an amount of the liquid to be supplied to the nozzle hole when the liquid is to be discharged. The method includes supplying the liquid from the liquid chamber to the nozzle hole of the nozzle plate when the liquid is to be discharged and when the liquid is not to be discharged.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] A more complete appreciation of embodiments and the many attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings.

[0007] FIG. 1 is a schematic sectional view of a liquid discharge apparatus according to a first embodiment, illustrating its overall configuration.

[0008] FIG. 2 is a magnified view of a nozzle hole and elements around the nozzle hole, where the nozzle hole is almost completely closed by a valve.

[0009] FIG. 3 is a magnified view of a nozzle hole and elements around the nozzle hole, where the nozzle hole is not completely closed by a valve.

[0010] FIG. 4 is a block diagram illustrating a functional configuration of a controller provided for a liquid discharge apparatus, according to a first embodiment.

[0011] FIG. 5 is a diagram illustrating the driving voltage of a valve conveyor provided for a liquid discharge apparatus according to a first embodiment.

[0012] FIG. 6 is a diagram illustrating how an end face of a valve moves according to the driving voltage illustrated in FIG. 5.

[0013] FIG. 7 is a schematic perspective view of a valve provided for a liquid discharge apparatus according to a first modification of the first embodiment.

[0014] FIG. 8 is a schematic sectional view of a valve provided for a liquid discharge apparatus according to a second modification of the first embodiment.

[0015] FIG. 9 is a schematic sectional view of a valve provided for a liquid discharge apparatus according to a third modification of the first embodiment.

[0016] FIG. 10 is a schematic perspective view of a nozzle plate provided for a liquid discharge apparatus according to a fourth modification of the first embodiment.

[0017] FIG. 11 is a schematic sectional view of a nozzle plate provided for a liquid discharge apparatus according to a fifth modification of the first embodiment.

[0018] FIG. 12 is a schematic view of a nozzle plate provided for a liquid discharge apparatus according to a sixth modification of the first embodiment.

[0019] FIG. 13 is a schematic sectional view of a liquid discharge apparatus according to a second embodiment, where liquid is not discharged.

[0020] FIG. 14 is a schematic sectional view of a liquid discharge apparatus according to the second embodiment, where liquid is discharged.

[0021] FIG. 15 is a schematic diagram illustrating a configuration of an applicator according to a third embodiment.

[0022] FIG. 16 is a diagram illustrating a first example of the arrangement of an applicator on an object, according to a third embodiment.

[0023] FIG. 17 is a diagram illustrating a second example of the arrangement of an applicator on an object, according to the third embodiment.

[0024] FIG. 18 is a first schematic sectional view of a nozzle plate provided for a liquid discharge apparatus according to a seventh modification.

[0025] FIG. 19 is a second schematic sectional view of a nozzle plate provided for a liquid discharge apparatus according to a seventh modification.

[0026] The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted.

DETAILED DESCRIPTION

[0027] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to limit the present disclosure. As used herein, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes” and/or “including”, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0028] In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the present disclosure is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have the same structure, operate in a similar manner, and achieve a similar result.

[0029] In the following description, illustrative embodiments will be described with reference to acts and symbolic representations of operations (e.g., in the form of flowcharts) that may be implemented as program modules or functional processes including routines, programs, objects, components, data structures, etc., that perform particular tasks or implement particular abstract data types and may be implemented using existing hardware at existing network elements or control

nodes. Such existing hardware may include one or more central processing units (CPUs), digital signal processors (DSPs), application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), computers, or the like. These terms may be collectively referred to as processors. [0030] Unless specifically stated otherwise, or as is apparent from the discussion, terms such as “processing” or “computing” or “calculating” or “determining” or “displaying” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical, electronic quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

[0031] A liquid discharge apparatus and a method of discharging liquid are described below with reference to the accompanying drawings. A liquid discharge apparatus and a method of discharging a liquid according to embodiments of the present disclosure are described below to implement the technical ideas, and no limitation is indicated to the embodiments of the present disclosure given below. For example, the size, material, and shape of components and the relative positions of the arranged components are given by way of example in the following description, and the scope of the present disclosure is not limited thereto unless particularly specified. For example, the size of these elements and the relative positions of these elements may be exaggerated for purposes of illustration in the drawings. In the description given below with reference to the drawings, identical or similar names or reference signs denote like elements, and overlapping description may be simplified or omitted as appropriate.

[0032] The directions that the X-axis and the Y-axis indicate are applicable to the drawings referred to in the following description. A direction in which the X-axis extends is referred to as the X-axis direction, and a direction in which the Y-axis extends is referred to as the Y-axis direction. The direction that is indicated by an arrow in the X-axis direction is referred to as a +X-axis direction, and the direction opposite to the +X-axis direction is referred to as a -X-axis direction. The direction that is indicated by an arrow in the Y-axis direction is referred to as a +Y-axis direction, and the direction opposite to the +Y-axis direction is referred to as a -Y-axis direction. In the description given below, it is assumed that the direction of the normal to the nozzle plate provided for the liquid discharge apparatus is parallel to the Y-axis direction. These expressions that indicate specific directions are merely used to indicate a relation such as a relative position, orientation, and direction, and it is not necessary for those expressions to be consistent with the relation in use.

[0033] In order to avoid complication in the drawings, a schematic diagram in which the illustration of some parts or members is omitted may be used, or an end view that merely illustrates a sectional view may be used. The parallelism may include an error of ± 10 degrees or less with respect to 0 degrees. The orthogonality may include an error of ± 10 degrees or less with respect to 90 degrees. The arrangement is not limited to a case of direct contact, and includes a case of indirect disposition such as contact through another member.

First Embodiment

[0034] The configuration or structure of the liquid discharge apparatus **100** according to the first embodiment is described below with reference to FIG. **1** to FIG. **3**.

[0035] FIG. **1** is a schematic sectional view of the liquid discharge apparatus **100** according to the first embodiment, illustrating its overall configuration.

[0036] FIG. **1** illustrates a cross section of the liquid discharge apparatus **100** including a plurality of nozzle holes **101** provided for the liquid discharge apparatus **100**.

[0037] FIG. **2** is a magnified view of the nozzle hole **101** and elements around the nozzle hole **101**, where the nozzle hole **101** is almost completely closed by the valve **104**.

[0038] FIG. **3** is a magnified view of the of nozzle hole **101** and elements around the nozzle hole **101**, where the nozzle hole **101** is not completely closed by the valve **104**.

[0039] FIG. **2** and FIG. **3** are magnified views of a region corresponding to the region II in FIG. **1**.

[0040] The liquid discharge apparatus **100** includes a liquid chamber **103** including a nozzle plate **102** having a nozzle hole **101** through which liquid is discharged, a supply unit **110** that supplies a pressurized liquid P to the liquid chamber **103**, a valve **104** provided for the liquid chamber **103**, and a valve conveyor **105** to move the valve **104**. As illustrated in FIG. **1**, the liquid discharge apparatus **100** includes a controller **120** that controls the movement of the valve **104** by the valve conveyor **105**, and a frame **106** to which an end of the valve conveyor **105** that does not face the valve **104** is coupled. The liquid discharge apparatus **100** further includes a supply port **107** through which the liquid P is supplied from the supply unit **110** to the liquid chamber **103**, and a discharge port **108** through which the liquid P in the liquid chamber **103** is discharged to the supply unit **110**. [0041] In the liquid discharge apparatus **100**, the pressurized liquid P that is supplied to the liquid chamber **103** by the supply unit **110** is discharged as the nozzle holes **101** formed on the liquid chamber **103** are opened by the movement of the valves **104**, and is not discharged when the nozzle holes **101** are closed by the movement of the valves **104**. The liquid discharge apparatus **100** is a liquid discharge apparatus in which valves open and close. In the liquid discharge apparatus **100**, the liquid P is discharged through the nozzle hole **101** from the inside to the outside of the liquid chamber **103** in a direction parallel to the Y-axis direction or a direction of the normal to the nozzle plate **102**.

[0042] As illustrated in FIG. **1**, the nozzle holes **101** are arrayed on the nozzle plate **102** in the X-axis direction. The liquid chambers **103**, which correspond to the nozzle holes **101**, are linked to each other in the X-axis direction to form one common chamber. The liquid P that is supplied from the supply unit **110** flows through the common chamber composed of the liquid chambers **103**. The liquid discharge apparatus **100** discharges the liquid P flowing in the common chamber from the nozzle holes **101** on an individual basis.

[0043] In such liquid discharge apparatuses where valves open and close, as illustrated in FIG. **2**, if an end **101a** of the nozzle hole **101** facing the valve **104** is blocked and closed by the valve **104** when the liquid P is not to be discharged, the liquid P may remain inside the nozzle hole **101**. The liquid P in the nozzle hole **101** illustrated in FIG. **2** is the liquid remaining inside the nozzle hole **101**. In addition to the end **101a**, the nozzle hole **101** has an end **101b** on the other side that is open to the atmosphere. For this reason, when the nozzle hole **101** is left for a long time, the liquid P that remains in the nozzle hole **101** is exposed to the atmosphere and is dried. Due to such drying, the liquid P in the nozzle hole **101** may be thickened or may adhere to, for example, the inner wall of the nozzle hole **101**. Due to such thickening of the liquid P or the adhesion of the liquid P to the nozzle hole **101**, when the liquid P is discharged from the nozzle hole **101** for the next time, discharge abnormality may take place in which the liquid P is not discharged or the liquid P is turned and discharged.

[0044] For example, in the related art, in order to reduce the chances of abnormal discharge, vibration is given to the liquid levels inside the nozzle holes when liquid is not to be discharged. By so doing, the thickened liquid is stirred in the liquid chamber. However, in such liquid discharge apparatuses where valves open and close, the liquid inside the liquid chamber is pressurized. For this reason, when liquid is not to be discharged, vibration cannot be given to the liquid levels inside the nozzle holes. For this reason, in such liquid discharge apparatuses where valves open and close, the thickened liquid cannot be stirred in the liquid chamber, and discharge abnormality cannot be reduced.

[0045] In the liquid discharge apparatus **100**, when the liquid P is to be discharged and when the liquid P is not to be discharged, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101**. The amount of the liquid P to be supplied to the nozzle holes **101** when the liquid P is not to be discharged is smaller than the amount of the liquid P to be supplied to the nozzle holes **101** when the liquid P is to be discharged.

[0046] For example, as illustrated in FIG. **3**, in the liquid discharge apparatus **100**, even when the liquid P is not discharged, the end **101a** of the nozzle hole **101** is not completely closed by the end

face **104a** of the valve **104** that face the nozzle plate **102** such that the gap **G0** appears between the valve **104** and the nozzle plate **102**. As the gap **G0** is made, the liquid **P** can flow through the gap **G0**, and the liquid **P** is supplied from the liquid chamber **103** to the nozzle holes **101**. From another viewpoint, in the liquid discharge apparatus **100**, the gap **G0** is made to decrease the degree of hermetically sealing of the nozzle hole **101** by the valve **104**, and the liquid **P** can be supplied to the nozzle holes **101** even when the liquid **P** is not to be discharged.

[0047] The gap **G0** when the liquid **P** is not to be discharged is narrower than the gap between the valve **104** and the nozzle plate **102** when the liquid **P** is to be discharged. Accordingly, the amount of the liquid **P** to be supplied to the nozzle holes **101** when the liquid **P** is not to be discharged becomes smaller than the amount of the liquid **P** to be supplied to the nozzle holes **101** when the liquid **P** is to be discharged, and the liquid **P** is not discharged from the nozzle holes **101**.

[0048] In the liquid discharge apparatus **100**, when the liquid **P** is not to be discharged, the liquid **P** is supplied from the liquid chamber **103** to the nozzle holes **101**, but the liquid **P** is not discharged from the nozzle holes **101**. Accordingly, when the liquid **P** remaining in the nozzle holes **101** is exposed to the outside air and dried, the liquid **P** that is fresh is supplied from the liquid chamber **103** to the nozzle holes **101**. Accordingly, the chances of the liquid **P**, which remains in the nozzle holes **101**, thickening or adhering to, for example, the inner walls of the nozzle holes **101** can be reduced. Accordingly, when the liquid **P** is discharged from the nozzle holes **101** again after a non-discharge period, the chances of discharge abnormality in which the liquid **P** is not discharged or the liquid **P** is turned and discharged can be reduced. When the liquid **P** is not discharged from the nozzle holes **101** in the present disclosure, the liquid **P** does not drip or leak out from the nozzle holes **101**.

[0049] In the liquid discharge apparatus **100**, as illustrated in FIG. 3, when the liquid **P** is not to be discharged, a part of the liquid **P** disposed in the nozzle hole **101** protrudes from the nozzle plate **102**. Due to such a configuration, when the liquid **P** is not to be discharged, the liquid **P** is supplied from the liquid chamber **103** to the nozzle holes **101**, but the liquid **P** is not discharged from the nozzle holes **101**.

[0050] In the liquid discharge apparatus **100**, the amount of the liquid **P** to be supplied to the nozzle holes **101** when the liquid **P** is not to be discharged varies depending on the physical properties of the liquid **P** or the environment near the nozzle holes. For example, the physical properties of the liquid **P** the viscosity or surface tension of the liquid **P** or the boiling point of the liquid **P**, and the environment near the nozzle holes is steam pressure near the nozzle holes. The amount of the liquid **P** to be supplied to the nozzle holes **101** when the liquid **P** is not to be discharged is changed according to the physical properties of the liquid **P**. Due to such a configuration, when the liquid **P** is not to be discharged, the liquid **P** is supplied from the liquid chamber **103** to the nozzle holes **101**, but the liquid **P** is not discharged from the nozzle holes **101**. Such a condition is achieved with high stability regardless of the physical properties of the liquid **P**. The amount of the liquid **P** to be supplied to the nozzle holes **101** when the liquid **P** is not to be discharged is changed according to the steam pressure near the nozzle holes. Due to such a configuration, when the liquid **P** is not to be discharged, the liquid **P** is supplied from the liquid chamber **103** to the nozzle holes **101**, but the liquid **P** is not discharged from the nozzle holes **101**. In a similar manner to the above, such a condition is achieved with high stability regardless of the environment near the nozzle holes.

[0051] Some components of the liquid discharge apparatus **100** or elements related to those components are described below in detail.

[0052] The liquid **P** is, for example, a liquid whose viscosity is greater than 10 millipascal-second (mPa·Math.s), a liquid whose TI value indicating mechanical viscosity is greater than 1.3, a liquid whose solid matter is greater than 20 weight (wt) %, or a liquid that contains particles whose particle diameter is wider than 5 micrometers (μm). For example, particles contained in the liquid **P** are active-material particles such as ceramic particles, carbon particles, and lithium transition metal oxide.

[0053] The nozzle plate **102** is a plate-like component including, for example, a metallic material. The nozzle plate **102** makes up a part of the liquid chamber **103**, and is arranged so as to face ends of the valves **104**. The nozzle hole **101** is a through hole formed on the nozzle plate **102** to penetrate in a direction of the normal to the nozzle plate **102**. The top face **102a** of the nozzle plate **102** is a top face of the nozzle plate **102** that faces the valve **104**.

[0054] The valves **104** are a plurality of stick-like members each of which extends in the Y-axis direction. The ends of the valves **104** on the +Y side are arranged so as to be accommodated inside the liquid chamber **103**. The valve **104** includes, for example, a metallic material. An end of the valve **104** that faces the nozzle plate **102** may be prepared as a member separate from the body of the valve **104**, and the separate member and the body of the valve **104** may be bonded together to make up the valve **104**.

[0055] The multiple valves **104** are paired with the nozzle holes **101** on a one-on-one basis, and are movable in the Y-axis direction. A part of each one of the multiple valves **104** on the +Y side is inserted into the liquid chamber **103**. The valve **104** opens and closes the nozzle hole **101** by the movement of the valve **104**. The valve conveyor **105** moves the valve **104** in the Y-axis direction according to the applied voltage. Due to such a configuration, it is switchable between a state in which the nozzle hole **101** is open as the end face **104a** of the valve **104** does not block the end **101a** of the nozzle hole **101** and a state in which the nozzle hole **101** is closed as the end face **104a** of the valve **104** blocks the end **101a** of the nozzle hole **101**.

[0056] The valve conveyor **105** includes a piezoelectric element that expands and contracts in response to an applied voltage. Such a piezoelectric element of the valve conveyor **105** may be, for example, a vertically-displacing piezoelectric element of d33 mode. The piezoelectric element of the valve conveyor **105** expands and contracts in the Y-axis direction in response to the applied voltages. The material of the piezoelectric element is, for example, lead zirconate titanate (PZT). The piezoelectric element of the valve conveyor **105** is a multi-layered piezoelectric element in which a plurality of piezoelectric elements are stacked on top of each other in the Y-axis direction in order to increase the amount of displacement caused by the expansion and contraction. The valve conveyor **105** illustrated in FIG. 1 moves the valve **104** in the Y-axis direction as the piezoelectric element expands and contracts in accordance with the applied voltage.

[0057] As the piezoelectric element of the valve conveyor **105** contract in the Y-axis direction in response to the applied voltages, the valve **104** moves in the -Y-axis direction, and the end face **104a** of the valve **104** is separated from the end **101a** of the nozzle hole **101** to open the nozzle hole **101**. As the piezoelectric element in the valve conveyor **105** expands in the Y-axis direction in response to the applied voltages, the valve **104** moves in the +Y-axis direction and the end face **104a** of the valve **104** blocks the end **101a** of the nozzle hole **101** to close the nozzle hole **101**.

[0058] As illustrated in FIG. 1, the supply unit **110** includes a liquid storage **111**, an air compressor **112**, an air tank **113**, a regulator **114**, and an air filter **115**. The supply unit **110** keeps the compressed air produced by the air compressor **112** in the air tank **113** on a temporary basis to maintain the pressure. Subsequently, the regulator **114** controls the supply unit **110** to reduce the pressure of the compressed air to a level of pressure required to discharge the liquid P, and controls the air filter **115** to remove, for example, foreign matter, moisture, or oil content. After that, the regulator **114** controls the supply unit **110** to increase the pressure inside the liquid storage **111**. The liquid P whose pressure is increased by the supply unit **110** is supplied to the liquid chamber **103** through the supply port **107**.

[0059] As the liquid P is pressurized by the supply unit **110**, the liquid P can be discharged to the outside of the liquid chamber **103** through the nozzle hole **101** when the nozzle hole **101** is opened by the valve **104**. The liquid P is not discharged when the nozzle hole **101** is closed by the valve **104**.

[0060] The valve conveyor **105** illustrated in FIG. 1 includes a piezoelectric element. However, no limitation is indicated thereby. For example, the valve conveyor **105** may include at least one of an

electromagnet that generates a magnetic field in response to an applied voltage, an air cylinder that converts compressed-air energy into a linear motion in response to an applied voltage, a motor actuator that is driven in response to an applied voltage, or a hydraulic mechanism that generates a hydraulic pressure in response to an applied voltage.

[0061] The liquid discharge apparatus **100** may be provided with the valve conveyor **105** that includes a piezoelectric element of d31 mode. The relation between the magnitude of the applied voltage and the expansion and contraction when such a piezoelectric element of d31 mode is adopted is completely different from the relation when a piezoelectric element of d33 mode is adopted. For example, the valve conveyor **105** may have the nozzle hole **101** at one end, and may include a hollow nozzle housing that has an inlet through which the liquid P is injected near the nozzle hole **101**, a piezoelectric element that is built in the nozzle housing and expands and contracts in response to the external application of a driving voltage, a valve that opens and closes a discharge orifice, a reverse spring mechanism arranged between the valve and the piezoelectric element, a sealing member fit to the valve to prevent the ink from flowing into the area on the piezoelectric element side, and a pair of lead wires coupled to the electrodes of the piezoelectric element to apply a voltage.

[0062] FIG. **4** is a block diagram illustrating a functional configuration of the controller **120**.

[0063] FIG. **5** is a diagram illustrating the driving voltage of the valve conveyor **105**.

[0064] FIG. **6** is a diagram illustrating how the end face **104a** of the valve **104** moves according to the driving voltage illustrated in FIG. **5**.

[0065] In FIG. **5** and FIG. **6**, the horizontal axis indicates time. In FIG. **5**, the vertical axis indicates the voltage value of the driving voltage. The vertical axis in FIG. **6** indicates the position of the end face **104a** of the valve **104** in the Y-axis direction. In FIG. **5** and FIG. **6**, the times on the horizontal axis are matched. The position of the valve **104** illustrated in FIG. **6** varies according to the voltage value of the driving voltage illustrated in FIG. **5**. In FIG. **5** and FIG. **6**, the voltage value of the driving voltage and the position of the end face **104a** have a substantially linear relation. Such a substantially linear relation indicates that those factors are in a linear relation as a whole, and includes a time-lag of the positional change of the end face **104a** in response to application of driving voltage and deviations from complete linearity such as errors in linearity and hysteresis.

[0066] The controller **120** illustrated in FIG. **4** includes an input unit **121**, a driving-voltage generation unit **122**, an amplifier **123**, and an output unit **124**. Such functions may be implemented by an electric circuit, or some of or all of those functions may be implemented by software or a central processing unit (CPU). These functions may be implemented by a plurality of circuits or a plurality of CPUs. The controller **120** generates, according to data input by the driving-voltage generation unit **122** through the input unit **121**, a driving voltage to drive the valve conveyor **105** to open and close the nozzle hole **101** by the valve **104**. The controller **120** amplifies the driving voltage generated by the driving-voltage generation unit **122** by the amplifier **123**, and then outputs the amplified driving voltage to the valve conveyor **105** through the output unit **124**.

[0067] The controller **120** outputs the waveform of the driving voltage, which is generated by the driving-voltage generation unit **122** and amplified by the amplifier **123**, to the liquid discharge apparatus **100**. By so doing, the driving voltage that varies over time is supplied to the valve conveyor **105**.

[0068] In the liquid discharge apparatus **100**, the controller **120** uses the movement of the valve **104** by the valve conveyor **105** to switch between a first mode used to discharge the liquid P and a second mode used not to discharge the liquid P. In the first mode, the end face **104a** of the valve **104** is separate the top face **102a** of the nozzle plate **102** by a first distance D1. A position Q2 in FIG. **6** corresponds to the position of the top face **102a** of the nozzle plate **102** in the Y-axis direction. In the second mode, the end face **104a** of the valve **104** is separate from the top face **102a** of the nozzle plate **102** by a second distance D2 shorter than the first distance D1. In the second mode of the liquid discharge apparatus **100**, the liquid P is supplied from the liquid chamber

103 to the nozzle holes **101** through the gap between the end face **104a** of the valve **104** and the top face **102a** of the nozzle plate **102**.

[0069] More specific description is given below. When a first voltage E1 is applied to the valve conveyor **105**, the piezoelectric element of the valve conveyor **105** contracts, and the end face **104a** of the valve **104** moves in the -Y-axis direction. Accordingly, the end face **104a** is separated from the end **101a** of the nozzle hole **101** to open the nozzle hole **101**. As the nozzle hole **101** opens, the liquid P is discharged from the nozzle holes **101**. When a third voltage E3, which is higher than the first voltage E1, is applied to the valve conveyor **105**, the piezoelectric element of the valve conveyor **105** expands, and the end face **104a** of the valve **104** moves in the +Y-axis direction. As a result, the end face **104a** of the valve **104** almost completely blocks the end **101a** of the nozzle hole **101** to close the nozzle hole **101**. As the nozzle hole **101** is closed, the liquid P is not discharged.

[0070] In the second mode of the liquid discharge apparatus **100**, a second voltage E2, which is greater than the first voltage E1 and smaller than the third voltage E3, is applied to the valve conveyor **105**. In such cases, the end face **104a** of the valve **104** does not completely block the end **101a** of the nozzle hole **101**, and the gap G0 appears as illustrated in FIG. 3. As the nozzle hole **101** is closed, the liquid P is not discharged. In other words, in the second mode, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101**, but the liquid P is not discharged from the nozzle holes **101**. In the second mode of the liquid discharge apparatus **100**, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101** through the gap G0 between the end face **104a** of the valve **104** and the top face **102a** of the nozzle plate **102**.

[0071] It is desired that the second distance D2 be shorter as the degree of the pressure applied to the liquid P is higher. When the length of the gap G0 in the Y-axis direction is consistent, the amount of the liquid P supplied from the liquid chamber **103** to the nozzle holes **101** increases as the degree of the pressure applied to the liquid P is higher. When the amount of the liquid P supplied from the liquid chamber **103** to the nozzle holes **101** increases to an excessive degree, the liquid P may leak out or drip from the nozzle holes **101**. The second distance D2 is shortened as the degree of the pressure applied to the liquid P is higher, so as to prevent the liquid P from leaking out or drips from the nozzle holes **101**.

[0072] Some modifications of the liquid discharge apparatus **100** are described below.

First Modification

[0073] FIG. 7 is a schematic perspective view of the valve **104** provided for the liquid discharge apparatus **100** according to a first modification of the first embodiment.

[0074] More specifically, FIG. 7 illustrates the end face **104a** of the valve **104** and the nozzle hole **101** and nozzle plate **102** that face the end face **104a**.

[0075] In the liquid discharge apparatus **100** according to the first modification, the first groove **41** is arranged on the end face **104a** of the valve **104**. In the liquid discharge apparatus **100** according to the first modification, when the liquid P is not discharged, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101** through the first groove **41**. These points are different from the liquid discharge apparatus **100** according to the first embodiment.

[0076] As the first groove **41** is arranged on the end face **104a** of the valve **104**, the nozzle hole **101** is blocked by the end face **104a** of the valve **104**. Even if the nozzle hole **101** is closed and the gap G0 in FIG. 3 is not present between the valve **104** and the nozzle plate **102**, the liquid P in the liquid chamber **103** can flow to the nozzle hole **101** through the first groove **41**. In other words, even if an end of the valve **104** is at a position Q2, the liquid P in the liquid chamber **103** can flow to the nozzle hole **101** through the first groove **41**. Due to such a configuration, in the liquid discharge apparatus **100** according to the first modification, when the liquid P is not to be discharged, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101**, but the liquid P is not discharged from the nozzle holes **101**. When the liquid P remaining in the nozzle holes **101** is exposed to the outside air and dried, the liquid P that is fresh is supplied from the liquid chamber **103** to the nozzle holes **101**. Accordingly, the chances of the liquid P, which

remains in the nozzle holes **101**, thickening or adhering to, for example, the inner walls of the nozzle holes **101** can be reduced. As described above, when the liquid P is discharged from the nozzle holes **101** again after a non-discharge period, the chances of discharge abnormality in which the liquid P is not discharged or the liquid P is turned and discharged can be reduced.

[0077] In the liquid discharge apparatus **100** according to the first modification, when the liquid P is not to be discharged, the gap **G0** in FIG. **3** may be present. In other words, it is not necessary for the end of the valve **104** to be at the position **Q2**, and the end of the valve **104** may be displaced from the position **Q2**. The first groove **41** is not limited to the groove that extends in one direction as illustrated in FIG. **7**. For example, the first groove **41** may include a plurality of grooves that intersect with each other, or may include a plurality of grooves extending radially from the center of the end face **104a**. The direction in which the liquid P is discharged from the nozzle hole **101** may vary depending on the shape of the first groove **41**. For this reason, from the viewpoint of reducing the influence of the shape of the first groove **41** on the direction in which the liquid P is discharged from the nozzle hole **101**, it is desired that the first groove **41** has point symmetry with respect to the center of the end face **104a**.

Second Modification

[0078] FIG. **8** is a schematic sectional view of the valve **104** provided for the liquid discharge apparatus **100** according to a second modification of the first embodiment.

[0079] FIG. **8** is a magnified view of a region corresponding to the region II in FIG. **1**.

[0080] In the liquid discharge apparatus **100** according to the second modification, first convex portions **42** are arranged on the end face **104a** of the valve **104**. In the liquid discharge apparatus **100** according to the second modification, when the liquid P is not discharged, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101** through a first gap **G1** formed as the first convex portion **42** contacts the nozzle plate **102**. These points are different from the liquid discharge apparatus **100** according to the first embodiment.

[0081] As the first convex portions **42** are arranged on the end face **104a** of the valve **104**, the nozzle hole **101** is blocked by the end face **104a** of the valve **104**, and the nozzle hole **101** is closed. In other words, even when the gap **G0** in FIG. **3** is absent, the liquid P in the liquid chamber **103** can flow to the nozzle hole **101** through the first gap **G1**. In other words, even if an end of the valve **104** is at the position **Q2**, the liquid P in the liquid chamber **103** can flow to the nozzle hole **101** through the first gap **G1**. Due to such a configuration, in the liquid discharge apparatus **100** according to the second modification, when the liquid P is not to be discharged, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101**, but the liquid P is not discharged from the nozzle holes **101**. When the liquid P remaining in the nozzle holes **101** is exposed to the outside air and dried, the liquid P that is fresh is supplied from the liquid chamber **103** to the nozzle holes **101**. Accordingly, the chances of the liquid P, which remains in the nozzle holes **101**, thickening or adhering to, for example, the inner walls of the nozzle holes **101** can be reduced. As described above, when the liquid P is discharged from the nozzle holes **101** again after a non-discharge period, the chances of discharge abnormality in which the liquid P is not discharged or the liquid P is turned and discharged can be reduced.

[0082] In the liquid discharge apparatus **100** according to the second modification, when the liquid P is not to be discharged, the gap **G0** in FIG. **3** may be present. In the liquid discharge apparatus **100** according to the second modification, when the liquid P is not discharged, the amount of the liquid P supplied from the liquid chamber **103** to the nozzle holes **101** may be changed depending on the shape of the first convex portions **42** or the positions at which the first convex portions **42** are arranged.

[0083] A shape that corresponds to the first convex portion **42** may be formed on a metal mold, and the valve **104** may be molded using the above metal mold to form the first convex portion **42**. A concave portion may be formed on the valve **104** by, for example, cutting, sandblasting, etching, and embossing, to form the first convex portion **42**. Alternatively, a member that makes up the first

convex portion **42** may be fixed to the end face **104a** of the valve **104** to arrange the first convex portion **42**. Alternatively, hard particles such as aluminum oxide may be spread out onto the end face **104a** of the valve **104** and then such spread hard particles may be coated to arrange the first convex portion **42**. Further, the valve **104** may be formed of an elastic material, and hard particles such as aluminum oxide may be strongly pressed against the valve **104** for fixation to arrange the first convex portion **42**.

Third Modification

[0084] FIG. **9** is a schematic sectional view of the valve **104** provided for the liquid discharge apparatus **100** according to a third modification of the first embodiment.

[0085] FIG. **9** is a magnified view of a region corresponding to the region II in FIG. **1**.

[0086] In the liquid discharge apparatus **100** according to the third modification, the end face **104a** of the valve **104** has a first porous structure **43**. In the liquid discharge apparatus **100** according to the third modification, when the liquid P is not to be discharged, the liquid P is supplied to the nozzle holes **101** through a pore of the first porous structure **43**. These points are different from the liquid discharge apparatus **100** according to the first embodiment.

[0087] As the end face **104a** of the valve **104** has the first porous structure **43**, the nozzle hole **101** is blocked by the end face **104a** of the valve **104**, and the nozzle hole **101** is closed. In other words, even when the gap G0 in FIG. **3** is absent, the liquid P in the liquid chamber **103** can flow to the nozzle hole **101** through a pore of the first porous structure **43**. In other words, even if an end of the valve **104** is at the position Q2, the liquid P in the liquid chamber **103** can flow to the nozzle hole **101** through a pore of the first porous structure **43**. Due to such a configuration, in the liquid discharge apparatus **100** according to the third modification, when the liquid P is not to be discharged, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101**, but the liquid P is not discharged from the nozzle holes **101**. When the liquid P remaining in the nozzle holes **101** is exposed to the outside air and dried, the liquid P that is fresh is supplied from the liquid chamber **103** to the nozzle holes **101**. Accordingly, the chances of the liquid P, which remains in the nozzle holes **101**, thickening or adhering to, for example, the inner walls of the nozzle holes **101** can be reduced. As described above, when the liquid P is discharged from the nozzle holes **101** again after a non-discharge period, the chances of discharge abnormality in which the liquid P is not discharged or the liquid P is turned and discharged can be reduced.

[0088] In the liquid discharge apparatus **100** according to the third modification, when the liquid P is not to be discharged, the gap G0 in FIG. **3** may be present. The first porous structure **43** may be formed of, for example, metallic material or resin material that has porous properties. A member that has the first porous structure **43** may be prepared as a separate body from the body of the valve **104**, and the body of the valve **104** and the member having the first porous structure **43** may be bonded together to form the valve **104**. The size of the pore of the first porous structure **43** may be changed to change the amount of the liquid P supplied from the liquid chamber **103** to the nozzle holes **101** when the liquid P is not discharged.

Fourth Modification

[0089] FIG. **10** is a schematic sectional view of the nozzle plate **102** provided for the liquid discharge apparatus **100** according to a fourth modification of the first embodiment.

[0090] More specifically, FIG. **10** illustrates the end face **104a** of the valve **104** and the nozzle hole **101** and nozzle plate **102** that face the end face **104a**.

[0091] In the liquid discharge apparatus **100** according to the fourth modification, a second groove **21** is arranged top face **102a** of the nozzle plate **102**, when viewed in a direction of the normal to the nozzle plate **102**, which is parallel to, for example, the Y-axis direction, at least some of the second groove **21** is at a position outside the valve **104**. In the liquid discharge apparatus **100** according to the fourth modification, when the liquid P is not to be discharged, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101** through the second groove **21**. These points are different from the liquid discharge apparatus **100** according to the first embodiment.

[0092] AS the second groove **21** is arranged top face **102a** of the nozzle plate **102**, the nozzle hole **101** is blocked by the end face **104a** of the valve **104**, and the nozzle hole **101** is closed. In other words, even when the gap **G0** in FIG. **3** is absent, the liquid P in the liquid chamber **103** can flow to the nozzle hole **101** through a pore of the first porous structure **43**. In other words, even if an end of the valve **104** is at the position **Q2**, the liquid P in the liquid chamber **103** can flow to the nozzle hole **101** through the second groove **21**. Due to such a configuration, in the liquid discharge apparatus **100** according to the fourth modification, when the liquid P is not to be discharged, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101**, but the liquid P is not discharged from the nozzle holes **101**. When the liquid P remaining in the nozzle holes **101** is exposed to the outside air and dried, the liquid P that is fresh is supplied from the liquid chamber **103** to the nozzle holes **101**. Accordingly, the chances of the liquid P, which remains in the nozzle holes **101**, thickening or adhering to, for example, the inner walls of the nozzle holes **101** can be reduced. As described above, when the liquid P is discharged from the nozzle holes **101** again after a non-discharge period, the chances of discharge abnormality in which the liquid P is not discharged or the liquid P is turned and discharged can be reduced.

[0093] In the liquid discharge apparatus **100** according to the fourth modification, when the liquid P is not to be discharged, the gap **G0** in FIG. **3** may be present. The second grooves **21** illustrated in FIG. **10** include two or more grooves that intersect with each other. However, no limitation is intended thereby, and the second groove **21** may include, for example, one groove that extends in one direction or a plurality of grooves that extend radially around the nozzle hole **101**. The direction in which the liquid P is discharged from the nozzle hole **101** may vary depending on the shape of the second groove **21**. For this reason, from the viewpoint of reducing the influence of the shape of the second groove **21** on the direction in which the liquid P is discharged from the nozzle hole **101**, it is desired that the second groove **21** has point symmetry with respect to the nozzle hole **101**.

Fifth Modification

[0094] FIG. **11** is a schematic sectional view of the nozzle plate **102** provided for the liquid discharge apparatus **100** according to a fifth modification of the first embodiment. FIG. **11** is a magnified view of a region corresponding to the region II in FIG. **1**.

[0095] In the liquid discharge apparatus **100** according to the fifth modification, the second convex portion **22** is arranged on a part of the top face **102a** of the nozzle plate **102**. In the liquid discharge apparatus **100** according to the fifth modification, when the liquid P is not discharged, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101** through a second gap **G2** formed as the second convex portion **22** contacts the valve **104**. These points are different from the liquid discharge apparatus **100** according to the first embodiment.

[0096] AS the second convex portion **22** is arranged on the top face **102a** of the nozzle plate **102**, the nozzle hole **101** is blocked by the end face **104a** of the valve **104**, and the nozzle hole **101** is closed. In other words, even when the gap **G0** in FIG. **3** is absent, the liquid P in the liquid chamber **103** can flow to the nozzle hole **101** through the second gap **G2**. In other words, even if an end of the valve **104** is at the position **Q2**, the liquid P in the liquid chamber **103** can flow to the nozzle hole **101** through the second gap **G2**. In the present embodiment, the second convex portion **22** contacts the valve **104** at the position **Q2**. Due to such a configuration, in the liquid discharge apparatus **100** according to the fifth modification, when the liquid P is not to be discharged, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101**, but the liquid P is not discharged from the nozzle holes **101**. When the liquid P remaining in the nozzle holes **101** is exposed to the outside air and dried, the liquid P that is fresh is supplied from the liquid chamber **103** to the nozzle holes **101**. Accordingly, the chances of the liquid P, which remains in the nozzle holes **101**, thickening or adhering to, for example, the inner walls of the nozzle holes **101** can be reduced. As described above, when the liquid P is discharged from the nozzle holes **101** again after a non-discharge period, the chances of discharge abnormality in which the liquid P is not discharged

or the liquid P is turned and discharged can be reduced.

[0097] In the liquid discharge apparatus **100** according to the fifth modification, when the liquid P is not to be discharged, the gap **G0** in FIG. **3** may be present. In the liquid discharge apparatus **100** according to the fifth modification, the amount of the liquid P supplied from the liquid chamber **103** to the nozzle holes **101** when the liquid P is not to be discharged may be changed depending on the shape or size of the second convex portions **22** or the positions at which the second convex portions **22** are arranged.

[0098] The second convex portion **22** can be formed on the nozzle plate **102** by, for example, cutting, sandblasting, etching, or embossing. Alternatively, a member that makes up the second convex portion **22** may be fixed to the top face **102a** of the nozzle plate **102** to arrange the second convex portion **22**. Alternatively, hard particles such as aluminum oxide are spread out onto the top face **102a** of the nozzle plate **102** and then such spread hard particles may be coated to arrange the second convex portion **22**. Further, the nozzle plate **102** may be formed of an elastic material, and hard particles such as aluminum oxide may be strongly pressed against the nozzle plate **102** for fixation to arrange the second convex portion **22**.

[0099] When the second convex portion **22** is arranged on a part of the top face **102a** of the nozzle plate **102**, it is desired that the second convex portion **22** be absent near the nozzle hole. In other words, it is desired that, on the top face **102a** of the nozzle plate **102**, an area free of the second convex portion **22** be arranged between the nozzle hole **101** and an area in which the second convex portion **22** is formed. As no convex portion is present near the nozzle hole, the roundness of the edge of the nozzle hole into which the liquid flows can be maintained, and the liquid can appropriately be discharged.

Sixth Modification

[0100] FIG. **12** is a schematic sectional view of the nozzle plate **102** provided for the liquid discharge apparatus **100** according to a sixth modification of the first embodiment.

[0101] FIG. **12** is a magnified view of a region corresponding to the region II in FIG. **1**.

[0102] The liquid discharge apparatus **100** according to the sixth modification has a second porous structure **23** on a part of the top face **102a** of the nozzle plate **102**. In the liquid discharge apparatus **100** according to the sixth modification, when the liquid P is not to be discharged, the liquid P is supplied to the nozzle holes **101** through a pore of the second porous structure **23**. These points are different from the liquid discharge apparatus **100** according to the first embodiment.

[0103] As the top face **102a** of the nozzle plate **102** has the second porous structure **23**, the nozzle hole **101** is blocked by the end face **104a** of the valve **104**, and the nozzle hole **101** is closed. In other words, even when the gap **G0** in FIG. **3** is absent, the liquid P in the liquid chamber **103** can flow to the nozzle hole **101** through a pore of the second porous structure **23**. In the present embodiment, the second porous structure **23** contacts the valve **104** at the position **Q2**. Due to such a configuration, in the liquid discharge apparatus **100** according to the sixth modification, when the liquid P is not to be discharged, the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101**, but the liquid P is not discharged from the nozzle holes **101**. When the liquid P remaining in the nozzle holes **101** is exposed to the outside air and dried, the liquid P that is fresh is supplied from the liquid chamber **103** to the nozzle holes **101**. Accordingly, the chances of the liquid P, which remains in the nozzle holes **101**, thickening or adhering to, for example, the inner walls of the nozzle holes **101** can be reduced. As described above, when the liquid P is discharged from the nozzle holes **101** again after a non-discharge period, the chances of discharge abnormality in which the liquid P is not discharged or the liquid P is turned and discharged can be reduced.

[0104] In the liquid discharge apparatus **100** according to the sixth modification, it is not always necessary for the gap **G0** in FIG. **3** to be absent when the liquid P is not to be discharged. In other words, it is not always necessary for the end of the valve **104** to be at the position **Q2**, and the gap **G0** in FIG. **3** may be present. The second porous structure **23** may be formed of, for example, metallic material or resin material that has porous properties. A member that has second porous

structure **23** may be prepared as a separate body from the body of the nozzle plate **102**, and the body of the nozzle plate **102** and the member having the second porous structure **23** may be bonded together to form the nozzle plate **102**. The size of the pore of the second porous structure **23** may be changed to change the amount of the liquid P supplied from the liquid chamber **103** to the nozzle holes **101** when the liquid P is not to be discharged.

Seventh Modification

[0105] When the valve conveyor **105** includes a piezoelectric element, the amount of movement of the valve **104** for the driving voltage per unit voltage may change depending on the temperature. When the total number of times the valve conveyor **105** has expanded and contracted increases, the amount of movement of the valve **104** for the driving voltage per unit voltage may decrease.

[0106] In the liquid discharge apparatus **100** according to the seventh modification, the second voltage E2 illustrated in FIG. 5 to be applied to the valve conveyor **105** when the liquid P is not to be discharged is corrected in accordance with the temperature of the valve conveyor **105** or the total number of times the valve conveyor **105** has expanded and contracted. Accordingly, the amount of the liquid P supplied to the nozzle holes **101** when the liquid P is not to be discharged is stabilized.

Eighth Modification

[0107] In the liquid discharge apparatus **100** according to the eighth modification, the controller **120** may further control the valve conveyor **105** to move the valve **104**, switching the first mode or the second mode to a third mode used not to supply the liquid P to the nozzle hole **101**. In the second mode, the end face **104a** of the valve **104** may be separate from the top face **102a** of the nozzle plate **102** by a third distance shorter than the second distance D2, and may contact the top face **102a** of the nozzle plate **102**.

[0108] If the liquid P is supplied from the liquid chamber **103** to the nozzle holes **101**, it may become difficult to clean, for example, the nozzle hole **101** or the nozzle plate **102**. In the liquid discharge apparatus **100** according to the eighth modification, when, for example, the nozzle hole **101** or the nozzle plate **102** is to be cleaned, the third mode is selected such that the liquid P is not supplied from the liquid chamber **103** to the nozzle holes **101**. This enables easy cleaning.

Ninth Modification

[0109] In the liquid discharge apparatus **100** according to the ninth modification, the controller **120** switches between the second mode and the third mode at prescribed time intervals or predetermined length of time. By switching between the second mode and the third mode at prescribed time intervals, the amount of the liquid P supplied from the liquid chamber **103** to the nozzle holes **101** when the liquid P is not to be discharged can be adjusted.

[0110] The degree of hermetically sealing of the nozzle hole **101** by the valve **104** may change due to, for example, the changes over time on the valve **104**, and the amount of the liquid P to be supplied from the liquid chamber **103** to the nozzle holes **101** may vary. In such cases, in the liquid discharge apparatus **100** according to the ninth modification, the mode is switched between the second mode and the third mode at prescribed time intervals to adjust the amount of the liquid P supplied from the liquid chamber **103** to the nozzle holes **101** when the liquid P is not to be discharged, in accordance with the changes in the degree of hermetically sealing of the nozzle hole **101** by the valve **104**. Accordingly, the amount of the liquid P supplied to the nozzle holes **101** when the liquid P is not to be discharged is stabilized. The amount of the liquid P to be supplied can be controlled by the length of time for which the distance between the end face **104a** and the top face **102a** is controlled to the second distance D2 or the time interval of the control. Due to such a configuration, the amount of the liquid P to be supplied can be controlled with a high degree of precision. The control of the gap by the driving voltage may have a relatively large variations in the amount of the liquid to be supplied. When the gaps are to be controlled by the driving voltage for multiple nozzles, fine adjustment for the voltage is to be performed for each nozzle, and the circuitry becomes complicated and expensive. By contrast, the adjustable of the time and interval

can be programmed, and detailed adjustment for the voltage is not necessary for each nozzle. Although the voltage is to be adjusted for each mode, simple circuitry and a low production cost can be achieved.

Tenth Modification

[0111] For example, as the steam pressure around the nozzle hole **101** is higher, the amount of evaporation from the nozzle hole **101** increases. In a similar manner, as the steam pressure around the nozzle hole **101** is lower, the amount of evaporation from the nozzle hole **101** decreases.

[0112] In the liquid discharge apparatus **100** according to the tenth modification, the controller **120** may control the amount of the liquid P supplied from the liquid chamber **103** to the nozzle holes **101** when the liquid P is not to be discharged to increase as the vapor pressure around the nozzle hole **101** is higher, and may control that amount to decrease as the steam pressure around the nozzle hole **101** is lower. Due to such a configuration, when the steam pressure around the nozzle hole **101** changes, the amount of the liquid P supplied from the liquid chamber **103** to the nozzle holes **101** when the liquid P is not to be discharged is maintained at a desired degree.

Other Modifications

[0113] When the valve conveyor **105** is fixed to the frame **106**, the distance between the end face **104a** of the valve **104** and the top face **102a** of the nozzle plate **102** may be made equivalent to the gap **G0**. By so doing, the gap **G0** can easily be obtained.

[0114] When the liquid P in which particles are dispersed is used, the second distance **D2** may be equal to or shorter than the maximum diameter of the particles, and it is desired that the second distance **D2** be equal to or shorter than the median diameter of the particles. For example, if particles flow from the liquid chamber **103** into the nozzle holes **101** when the liquid P is not to be discharged, such particles may adhere to the nozzle holes **101**, and the chances of discharge abnormality may increase. By making the second distance **D2** equal to or shorter than the maximum diameter of the particles, when the liquid P is not to be discharged, the particles can be prevented from flowing from the liquid chamber **103** into the nozzle holes **101**. Accordingly, discharge abnormality due to the adhesion of particles to the nozzle holes **101** can be reduced. When the second distance **D2** be equal to or shorter than the median diameter of the particles, discharge abnormality can further be reduced.

[0115] The liquid chamber **103** is not limited to the common chamber as illustrated in FIG. **1** where multiple liquid chambers are coupled to each other, and may be a plurality of individual liquid chambers that are separated from each other. The liquid discharge apparatus **100** is not limited to a liquid discharge apparatus having a plurality of nozzle holes **101**, and as will be described later in detail, may be a liquid discharge apparatus with a single nozzle hole **101**. Such a liquid discharge apparatus with a single nozzle hole may be referred to as a single-nozzle liquid discharge apparatus.

Second Embodiment

[0116] A liquid discharge apparatus **100a** according to a second embodiment is described below. In the description given below with reference to the drawings, identical or similar reference signs denote like elements, and overlapping description may be simplified or omitted as appropriate. The same condition applies to the embodiments described below.

[0117] The configuration or structure of a liquid discharge apparatus **100a** according to a second embodiment is described below with reference to FIG. **13** and FIG. **14**.

[0118] FIG. **13** is a schematic sectional view of the liquid discharge apparatus **100a** according to the second embodiment, where liquid P is not discharged.

[0119] FIG. **14** is a schematic sectional view of the liquid discharge apparatus **100a** according to the second embodiment, where liquid P is discharged.

[0120] The liquid discharge apparatus **100a** discharges the liquid P using a single nozzle. The liquid discharge apparatus **100a** includes a nozzle housing **4**, the nozzle hole **101**, the valve **104**, the valve conveyor **105**, a reverse spring mechanism **8**, a sealing member **6**, a pair of lead wires **9**,

and a lead wire **10**.

[0121] The nozzle housing **4** has the nozzle hole **101** through which the liquid P is discharged to the front end, and has an inlet **3** through which the liquid P is injected near the nozzle hole **101**. The valve conveyor **105** is built in the nozzle housing **4**, and expands and contracts in response to the external application of a driving voltage. The valve **104** opens and closes the nozzle hole **101**. The reverse spring mechanism **8** is arranged between the valve **104** and the valve conveyor **105**. The sealing member **6** is fitted on the valve **104** to prevent the liquid P from flowing into the area on the valve conveyor **105** side. The pair of lead wires **9** and the lead wire **10** are coupled to the electrodes of the valve conveyor **105** and used to apply a voltage.

[0122] As a whole, the nozzle housing **4** is formed in a cylindrical shape or an angularly cylindrical shape, and is closed except for the nozzle hole **101** and the inlet **3**. The nozzle hole **101** is an opening formed at a front end of the nozzle housing **4** through which the liquid P is discharged. The inlet **3** is arranged on a side face of the nozzle housing **4** near the nozzle hole **101**, and is coupled to a liquid storage. Moreover, through the inlet **3**, the liquid P is continuously supplied to liquid discharge apparatus **100a** by a biasing member.

[0123] The valve conveyor **105** is formed of, for example, zirconia ceramics, and is formed with an appropriate external shape and thickness in accordance with, for example, the amount of the liquid P to be discharged. The valve conveyor **105** is controlled by the driving voltage sent by the controller **120**.

[0124] The sealing member **6** is, for example, a packing, a gasket, or an O-ring, and the sealing member **6** is fitted on the valve **104** to prevent the ink from flowing into the area on the valve conveyor **105** side from the area on the inlet **3** side.

[0125] The reverse spring mechanism **8** is an elastic member formed by molding, for example, deformable rubber, soft resin, or thin metallic plate as desired. The reverse spring mechanism **8** includes a deforming portion **8a** whose cross section is approximately shaped like a trapezoid, a fixed portion **8b** fixed to the inner wall of the nozzle housing **4**, and a guide unit **8c** coupled to an end face of the valve conveyor **105**. The deforming portion **8a** is formed so as to contact a face of the valve **104** on its base end side. The long side, i.e., the base, of the deforming portion **8a** that is trapezoidal-shaped is a bent side **8d** coupled to the fixed portion **8b**.

[0126] When the driving voltage is applied to the valve conveyor **105**, the valve conveyor **105** expands. Accordingly, in the reverse spring mechanism **8**, the guide unit **8c** moves toward the nozzle hole **101** to press a portion of the bent side **8d** of the deforming portion **8a** around the center. As a result, the apex of the deforming portion **8a**, i.e., the top base of the trapezoid, which is coupled to the valve **104**, moves toward the valve conveyor **105**. The valve **104** is pulled toward the valve conveyor **105** by a distance d as illustrated in FIG. **13** to release the nozzle holes **101**.

[0127] The distance between the bent side **8d** and the apex of the deforming portion **8a** of the reverse spring mechanism **8**, which is a coupler with the valve **104**, is adjusted, or the length of the bent side **8d** is adjusted as desired. By so doing, the length of the movement of the valve **104** is made longer than the length of the expansion of the valve conveyor **105**. In other words, the reverse spring mechanism **8** can amplify a slight expansion of the valve conveyor **105**. Accordingly, the length of the valve conveyor **105**, which is expensive, can be shortened, and the production cost of the liquid discharge apparatus **100** can significantly be reduced. For example, the moving distance of the valve **104** may be made twice as much as the moving distance of an end face of the valve conveyor **105** to shorten the length of the valve conveyor **105**.

[0128] When no driving voltage is applied to the valve conveyor **105**, the valve conveyor **105** returns to its original shape. For this reason, no force is externally applied to the reverse spring mechanism **8**, and no deformation takes place. By contrast, when the driving voltage is applied to the valve conveyor **105**, the valve conveyor **105** expands, and the guide unit **8c** of the reverse spring mechanism **8** moves toward the nozzle hole **101** accordingly. As a result, the deforming portion **8a** deforms as if it is squeezed.

[0129] The liquid discharge apparatus **100** and the liquid discharging method are applicable to the liquid discharge apparatus **100a** with a single nozzle as described above with reference to FIG. **13** and FIG. **14**, and advantageous effects similar to those of the liquid discharge apparatus **100** and the liquid discharging method according to the first embodiment can be achieved.

Third Embodiment

[0130] FIG. **15** is a diagram illustrating a configuration of an applicator **1001** according to a third embodiment.

[0131] FIG. **16** is a diagram illustrating a first example of the arrangement of the applicator **1001** on an object U, according to the third embodiment.

[0132] FIG. **17** is a diagram illustrating a second example of the arrangement of the applicator **1001** on the object U, according to the third embodiment.

[0133] The applicator **1001** includes the liquid discharge apparatus **100**, a camera **1004** that is an example of an imaging device arranged near the liquid discharge apparatus **100**, an A-B table **1003** that moves the liquid discharge apparatus **100** and the camera **1004** in an A direction and a B direction, image and photo editing software S used to edit the images captured by the camera **1004**, and a controller **120g**. The controller **120g** controls the A-B table **1003** to operate based on a predetermined control program, and causes the liquid P to be discharged from the liquid discharge apparatus **100**. The applicator **1001** can apply the liquid P discharged from the liquid discharge apparatus **100** to the object U.

[0134] In the liquid discharge apparatus **100**, the liquid P is discharged from the nozzle holes **101** toward a surface of the object U to which liquid is to be applied. The liquid P is discharged from the nozzle holes **101** in a direction orthogonal to an A-B plane. The directions in which the liquid P is discharged from each of the nozzle holes **101** are parallel to each other. For example, the distance between the nozzle hole **101** and the surface of the object U to which liquid is to be applied is about 20 cm.

[0135] The A-B table **1003** include an A-axis **1005** formed with a linear movement mechanism and a B-axis **1006** that moves the A-axis **1005** in the Y-axis direction while holding the A-axis **1005** with two arms. The liquid discharge apparatus **100** and the camera **1004** are attached to a slider. The B-axis **1006** is provided with a shaft **1007**. The shaft **1007** is held by a robot arm **1008**. Due to such a configuration, the liquid discharge apparatus **100** can freely be arranged to any position with respect to the object U.

[0136] For example, when the object U is an automobile, the liquid discharge apparatus **100** may be arranged above the object U as illustrated in FIG. **16**, or the liquid discharge apparatus **100** may be arranged beside the object U as illustrated in FIG. **17**. The controller **120g** controls the operation of the robot arm **1008** based on a prescribed program.

[0137] The camera **1004** is mounted on the slider of the A-axis **1005** near the liquid discharge apparatus **100**, and captures, at certain minute intervals, an image of a prescribed range on the surface of the object U to which liquid is to be applied while moving in the A direction or the B direction. The camera **1004** is, for example, a digital camera. In the camera **1004**, the specification of, for example, a lens or resolution is selected as desired to enable the capturing of a plurality of subdivided images obtained by dividing the prescribed range of the surface to which liquid is to be applied. The camera **1004** continuously and automatically captures a plurality of subdivided images of the surface to which liquid is to be applied, in accordance with a program stored in the controller **120g** in advance.

[0138] As described above, the applicator **1001** includes the liquid discharge apparatus **100**. Due to such a configuration, for example, even if the distance between the object U and the nozzle hole **101** is long, the liquid P can be applied to a desired position of the object U with a high degree of precision. As the liquid discharge apparatus **100** can discharge the liquid P with high stability, the applicator **1001** can highly precisely coat the object U with the liquid P. The applicator **1001** may include the liquid discharge apparatus **100a** in place of the liquid discharge apparatus **100** or in

addition to the liquid discharge apparatus **100**.

[0139] The particles that the liquid P discharged by the applicator **1001** contains are, for example, aluminum flakes, mica, or titanium oxide. The liquid P that is discharged by the applicator **1001** is a liquid such as acetone and that easily evaporates, and when the concentration of the particles in the liquid remaining in the discharge orifice increases due to the evaporation while the liquid is not discharged, the stability when the liquid P is to be discharged again may decrease. With the adoption of a configuration or structure according to the embodiments of the present disclosure, the concentration of the liquid remaining in the discharge orifice can be prevented from increasing.

[0140] Note that numerous additional modifications and variations are possible in light of the above teachings. It is therefore to be understood that within the scope of the appended claims, the disclosure of the present disclosure may be practiced otherwise than as specifically described herein. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of this disclosure and appended claims.

[0141] FIG. **18** is a first schematic sectional view of the nozzle plate **102** provided for the liquid discharge apparatus **100** according to a seventh modification.

[0142] FIG. **19** is a second schematic sectional view of the nozzle plate **102** provided for the liquid discharge apparatus **100** according to the seventh modification.

[0143] On the top face of the nozzle plate **102** that faces the valve **104**, an area free of the second convex portion **22** may be arranged between the nozzle hole **101** and an area in which the second convex portion **22** is formed.

[0144] For example, the surface of the valve **104** at the front end may be made uneven by, for example, sandblasting, etching, and embossing. By so doing, channels for the liquid are formed. Alternatively, the surface of the nozzle plate **102** that faces or contacts at least the valve **104** may be made uneven by, for example, sandblasting, etching, and embossing. By so doing, channels for the liquid are formed.

[0145] The valve **104** may have a recessed end such that the inlet of the nozzle hole **101** and an area around that inlet do not contact such a recessed end of the valve **104**.

[0146] In an area B of the nozzle plate **102** that contacts the valve **104**, which is equal to or wider than an area A, uneven portions may be arranged by, for example, sandblasting, etching, and embossing. In an area C of the nozzle plate **102** near the inlet of the nozzle hole **101** and an area around that inlet, which does not contact a recessed end of the valve **104**, it is not necessary to arrange uneven portions. With these configurations, the roundness of the edge of the nozzle hole **101** into which the liquid flows can be maintained. Due to such a configuration, an inkjet printing head in which a valve with uneven portions, which serve as channels for the liquid, opens and closes can be implemented while maintaining the straightness in discharging liquid.

[0147] When the end face **104a** of the valve **104** is moved by the first distance DI to open the nozzle hole **101** in the liquid discharge apparatus according to the embodiments of the present disclosure, it is not always necessary for the nozzle hole **101** to be completely opened, and the nozzle hole **101** may partially be opened. Even if the nozzle hole **101** is partially opened, the chances of abnormally discharged can effectively be reduced. The amount of the liquid P to be discharged may be changed depending on how much the nozzle hole **101** is opened.

[0148] The numbers such as ordinal numbers and numerical values that indicates quantity are all given by way of example to describe the technologies to implement the embodiments of the present disclosure, and no limitation is indicated to the numbers given in the above description. The description as to how the elements are related to each other, coupled to each other, or connected to each other is given by way of example to describe the technologies to implement the embodiments of the present disclosure, and how the elements are related to each other, coupled to each other, or connected to each other to implement the functionality in the present disclosure is not limited thereby.

[0149] The division of blocks in the functional block diagrams is given by way of example. A plurality of blocks may be implemented as one block, or one block may be divided into a plurality of blocks. Alternatively, some functions may be moved to other blocks. The functions of a plurality of blocks that have similar functions may be processed in parallel or in a time-division manner by a single unit of hardware or software.

[0150] Each of the functions of the described embodiments of the present disclosure may be implemented by one or more processing circuits or circuitry. Processing circuitry includes a programmed processor, as a processor includes circuitry. A processing circuit also includes devices such as an application-specific integrated circuit (ASIC), digital signal processor (DSP), field-programmable gate array (FPGA), and conventional circuit components arranged to perform the recited functions. The processing circuit in the embodiments of the present disclosure as described above includes, for example, a device such as a processor that is programmed to execute software to implement functions, like a processor with electronic circuits, an application-specific integrated circuit (ASIC) that is designed to execute the above functions, a digital signal processor (DSP), a field-programmable gate array (FPGA), a complex programmable logic device (CPLD), and a circuit module known in the art.

[0151] With the application of the liquid discharge apparatus or the liquid discharging method according to the embodiments of the present disclosure, discharge abnormality can be reduced in liquid discharge apparatuses in which valves open and close and methods of discharging a liquid. Accordingly, the liquid discharge apparatus or the liquid discharging method according to the embodiments of the present disclosure is applicable to, for example, apparatuses for manufacturing elements such as electrochemical elements, printers with electrodes, painting or coating devices that apply discharged paint onto an object, or image forming apparatuses such as ink-jet printers as desired.

[0152] Typically, a secondary battery or an electrostatic capacitor that is an example of an electrochemical element may be used as an electrochemical element. In particular, the liquid discharge apparatus or the liquid discharging method according to the embodiments of the present disclosure is applicable to the manufacturing of lithium-ion secondary batteries. For example, with the application of the liquid discharge apparatus or the liquid discharging method according to the embodiments of the present disclosure, the period during which no liquid is discharged can significantly be extended compared with cases in which the liquid discharge apparatus and the liquid discharging method according to the embodiments of the present disclosure are not applied. More specifically, the period during which no liquid is discharged indicates a period between the time at which the discharging of the liquid P is stopped and the time at which the discharging of the liquid P starts again, and by definition, discharge abnormality does not occur when the discharging of the liquid P starts again. With the use of a rolled base to which the liquid discharge apparatus according to the embodiments of the present disclosure is applied, for example, when lithium-ion secondary batteries are manufactured, the chances of discharge abnormality can be reduced when the liquid is to be discharged again after the base whose length is 2000 m runs at 60 meters per minute (mpm) and thirty minutes have passed.

[0153] Aspects of the present disclosure are, for example, as follows.

First Aspect

[0154] A liquid discharge apparatus includes a liquid chamber including a nozzle plate having a nozzle hole, a supply unit to supply a liquid to the liquid chamber, a valve disposed in the liquid chamber, and a valve conveyor to move the valve. When the liquid is to be discharged and when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole. An amount of the liquid to be supplied to the nozzle hole when the liquid is not to be discharged is smaller than an amount of the liquid to be supplied to the nozzle hole when the liquid is to be discharged.

Second Aspect

[0155] In the liquid discharge apparatus according to the first aspect, the valve has an end face facing the nozzle plate, the end face has a first groove, and when the liquid is not discharged, the liquid is supplied from the liquid chamber to the nozzle hole through the first groove.

Third Aspect

[0156] In the liquid discharge apparatus according to the first aspect or the second aspect, the valve has an end face facing the nozzle plate, the end face has a first convex portion, and when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole through a first gap formed as the first convex portion contacts the nozzle plate.

Fourth Aspect

[0157] In the liquid discharge apparatus according to any one of the first aspect to the third aspect, the valve has an end face facing the nozzle plate, the end face has a first porous structure, and when the liquid is not to be discharged, the liquid is supplied to the nozzle hole through a pore of the first porous structure.

Fifth Aspect

[0158] In the liquid discharge apparatus according to any one of the first aspect to the fourth aspect, the nozzle plate has a top face facing the valve, wherein the top face has a second groove, and when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole through the second groove.

Sixth Aspect

[0159] In the liquid discharge apparatus according to any one of the first aspect to the fifth aspect, the nozzle plate has a top face facing the valve, the top face has a second convex portion, and when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole through a second gap formed as the second convex portion contacts the valve.

Seventh Aspect

[0160] In the liquid discharge apparatus according to the sixth aspect, the nozzle plate has a top face facing the valve, and the top face has an area free of the second convex portion between the nozzle hole and an area in which the second convex portion is formed.

Eighth Aspect

[0161] In the liquid discharge apparatus according to any one of the first aspect to the seventh aspect, the nozzle plate has a top face facing the valve, the top face has a second porous structure, and when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole through a pore of the second porous structure.

Ninth Aspect

[0162] The liquid discharge apparatus according to any one of the first aspect to the eighth aspect further includes a controller to control movement of the valve by the valve conveyor. The controller uses movement of the valve by the valve conveyor to switch between a first mode used to discharge the liquid and a second mode used not to discharge the liquid. In the first mode, the valve has an end face facing the nozzle plate, the nozzle plate has a top face facing the valve, and the end face of the valve is separate from the top face of the nozzle plate by a first distance. In the second mode, the end face of the valve is separate from the top face of the nozzle plate by a second distance shorter than the first distance. In the second mode, the liquid is supplied from the liquid chamber to the nozzle hole through a space between the end face of the valve and the top face of the nozzle plate.

Tenth Aspect

[0163] In the liquid discharge apparatus according to the ninth aspect, the controller controls the valve conveyor to move the valve, further switching the first mode or the second mode to a third mode used not to supply the liquid to the nozzle hole. In the third mode, the end face of the valve contacts the top face of the nozzle plate or is separated from the top face of the nozzle plate by a third distance shorter than the second distance.

Eleventh Aspect

[0164] In the liquid discharge apparatus according to the tenth aspect, the controller switches between the second mode and the third mode at prescribed time intervals.

Twelfth Aspect

[0165] A method of discharging a liquid uses a liquid discharge apparatus including a liquid chamber including a nozzle plate having a nozzle hole through which liquid is discharged, a supply unit to supply the pressurized liquid to the liquid chamber, a valve disposed in the liquid chamber, and a valve conveyor to move the valve. The liquid discharge apparatus supplies the liquid from the liquid chamber to the nozzle hole when the liquid is to be discharged and when the liquid is not to be discharged. An amount of the liquid supplied to the nozzle hole when the liquid is not to be discharged is smaller than an amount of the liquid supplied to the nozzle hole when the liquid is to be discharged.

Thirteenth Aspect

[0166] In the method of discharging the liquid according to the twelfth aspect, the amount of the liquid to be supplied to the nozzle hole when the liquid is not to be discharged varies depending on a type of the liquid.

Fourteenth Aspect

[0167] In the method of discharging the liquid according to the twelfth aspect or the thirteenth aspect, a controller controls movement of the valve by the valve conveyor, and the controller uses movement of the valve by the valve conveyor to switch between a first mode used to discharge the liquid and a second mode used not to discharge the liquid. In the first mode, the valve has an end face facing the nozzle plate, the nozzle plate has a top face facing the valve, and the end face of the valve is separate from the top face of the nozzle plate by a first distance. In the second mode, the end face of the valve is separate from the top face of the nozzle plate by a second distance shorter than the first distance. In the second mode, the liquid is supplied from the liquid chamber to the nozzle hole through a space between the end face of the valve and the top face of the nozzle plate.

Fifteenth Aspect

[0168] In the method of discharging the liquid according to the fourteenth aspect, particles are dispersed to the liquid, and the second distance is equal to or shorter than a maximum diameter of the particles.

Sixteenth Aspect

[0169] In the method of discharging the liquid according to the fourteenth aspect or the fifteenth aspect, the controller controls the valve conveyor to move the valve, further switching the first mode or the second mode to a third mode used not to supply the liquid to the nozzle hole. In the third mode, the end face of the valve is separated from the top face of the nozzle plate by a third distance shorter than the second distance or contacts the top face of the nozzle plate.

Seventeenth aspect

[0170] In the method of discharging the liquid according to the sixteenth aspect, the controller switches between the second mode and the third mode at prescribed time intervals.

Eighteenth Aspect

[0171] In the method of discharging the liquid according to any one of the twelfth aspect to the seventeenth aspect, when the liquid is not to be discharged, a part of the liquid disposed in the nozzle hole protrudes from the nozzle plate.

[0172] Any one of the above-described operations may be performed in various other ways, for example, in an order different from the one described above.

[0173] Each of the functions of the described embodiments may be implemented by one or more processing circuits or circuitry. Processing circuitry includes a programmed processor, as a processor includes circuitry. A processing circuit also includes devices such as an application-specific integrated circuit (ASIC), digital signal processor (DSP), field-programmable gate array (FPGA), and conventional circuit components arranged to perform the recited functions.

Claims

1. A liquid discharge apparatus comprising: a liquid chamber including a nozzle plate having a nozzle hole; a supply unit to supply a liquid to the liquid chamber; a valve disposed in the liquid chamber; and a valve conveyor to move the valve, wherein when the liquid is to be discharged and when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole, and wherein an amount of the liquid to be supplied to the nozzle hole when the liquid is not to be discharged is smaller than an amount of the liquid to be supplied to the nozzle hole when the liquid is to be discharged.
2. The liquid discharge apparatus according to claim 1, wherein the valve has an end face facing the nozzle plate, wherein the end face has a groove, and wherein when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole through the groove.
3. The liquid discharge apparatus according to claim 1, wherein the valve has an end face facing the nozzle plate, wherein the end face has a convex portion, and wherein when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole through a gap formed as the convex portion contacts the nozzle plate.
4. The liquid discharge apparatus according to claim 1, wherein the valve has an end face facing the nozzle plate, wherein the end face has a porous structure, and wherein when the liquid is not to be discharged, the liquid is supplied to the nozzle hole through a pore of the porous structure.
5. The liquid discharge apparatus according to claim 1, wherein the nozzle plate has a top face facing the valve, wherein the top face has a groove, and wherein when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole through the groove.
6. The liquid discharge apparatus according to claim 1, wherein the nozzle plate has a top face facing the valve, wherein the top face has a convex portion, and wherein when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole through a gap formed as the convex portion contacts the valve.
7. The liquid discharge apparatus according to claim 6, wherein the top face of the nozzle plate has an area free of the convex portion between the nozzle hole and an area in which the convex portion is formed.
8. The liquid discharge apparatus according to claim 1, wherein the nozzle plate has a top face facing the valve, wherein the top face has a porous structure, and wherein when the liquid is not to be discharged, the liquid is supplied from the liquid chamber to the nozzle hole through a pore of the porous structure.
9. The liquid discharge apparatus according to claim 1, further comprising a controller to control movement of the valve by the valve conveyor, wherein the controller uses movement of the valve by the valve conveyor to switch between a first mode used to discharge the liquid and a second mode used not to discharge the liquid, wherein in the first mode, the valve has an end face facing the nozzle plate, the nozzle plate has a top face facing the valve, and the end face of the valve is separate from the top face of the nozzle plate by a first distance, wherein in the second mode, the end face of the valve is separate from the top face of the nozzle plate by a second distance shorter than the first distance, and wherein in the second mode, the liquid is supplied from the liquid chamber to the nozzle hole through a space between the end face of the valve and the top face of the nozzle plate.
10. The liquid discharge apparatus according to claim 9, wherein the controller controls the valve conveyor to move the valve, further switching the first mode or the second mode to a third mode used not to supply the liquid to the nozzle hole, and wherein in the third mode, the end face of the valve contacts the top face of the nozzle plate or is separated from the top face of the nozzle plate by a third distance shorter than the second distance.
11. The liquid discharge apparatus according to claim 10, wherein the controller switches between

the second mode and the third mode at prescribed time intervals.

12. A method of discharging a liquid, the method comprising supplying the liquid from a liquid chamber to a nozzle hole of a nozzle plate when the liquid is to be discharged and when the liquid is not to be discharged, wherein an amount of the liquid to be supplied to the nozzle hole when the liquid is not to be discharged is smaller than an amount of the liquid to be supplied to the nozzle hole when the liquid is to be discharged.

13. The method according to claim 12, wherein the amount of the liquid to be supplied to the nozzle hole when the liquid is not to be discharged varies depending on a type of the liquid.

14. The method according to claim 12, further comprising: controlling movement of a valve disposed in the liquid chamber by a valve conveyor; and switching between a first mode used to discharge the liquid and a second mode used not to discharge the liquid, using the movement of the valve by the valve conveyor, wherein in the first mode, the valve has an end face facing the nozzle plate, the nozzle plate has a top face facing the valve, and the end face of the valve is separate from the top face of the nozzle plate by a first distance, wherein in the second mode, the end face of the valve is separate from the top face of the nozzle plate by a second distance shorter than the first distance, and wherein in the second mode, the liquid is supplied from the liquid chamber to the nozzle hole through a space between the end face of the valve and the top face of the nozzle plate.

15. The method according to claim 14, wherein particles are dispersed to the liquid, and wherein the second distance is equal to or shorter than a maximum diameter of the particles.

16. The method according to claim 14, further comprising controlling the valve conveyor to move the valve, further switching the first mode or the second mode to a third mode used not to supply the liquid to the nozzle hole, and wherein in the third mode, the end face of the valve is separated from the top face of the nozzle plate by a third distance shorter than the second distance or contacts the top face of the nozzle plate.

17. The method according to claim 16, further comprising switching between the second mode and the third mode at prescribed time intervals.

18. The method according to claim 12, wherein when the liquid is not to be discharged, a part of the liquid disposed in the nozzle hole protrudes from the nozzle plate.
